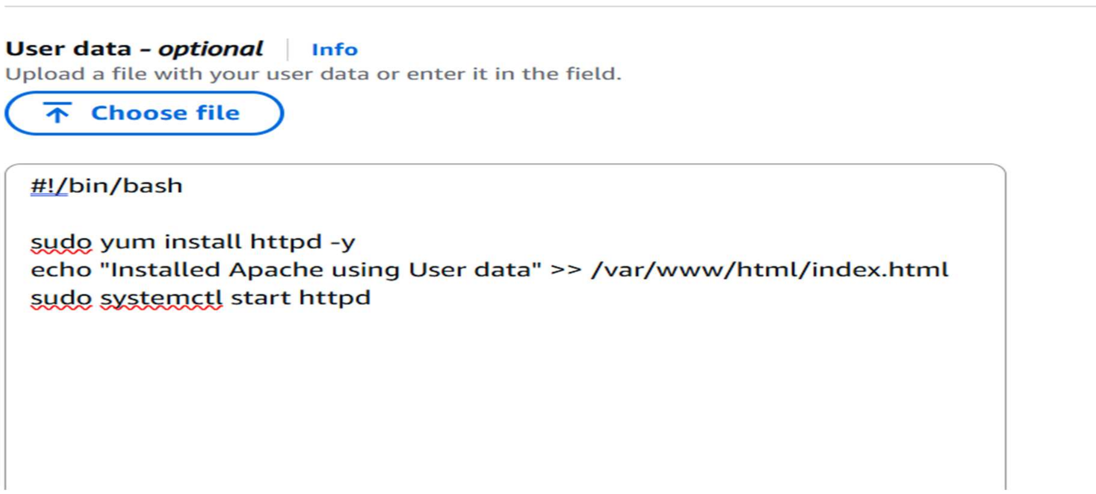
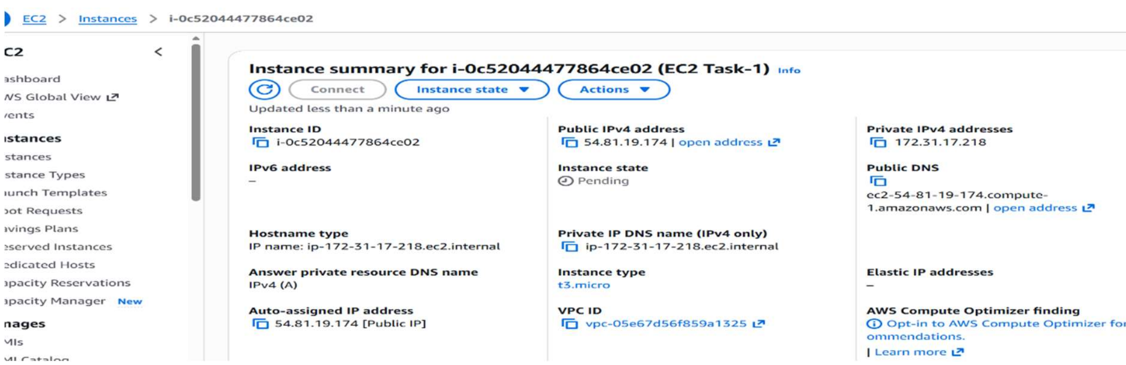
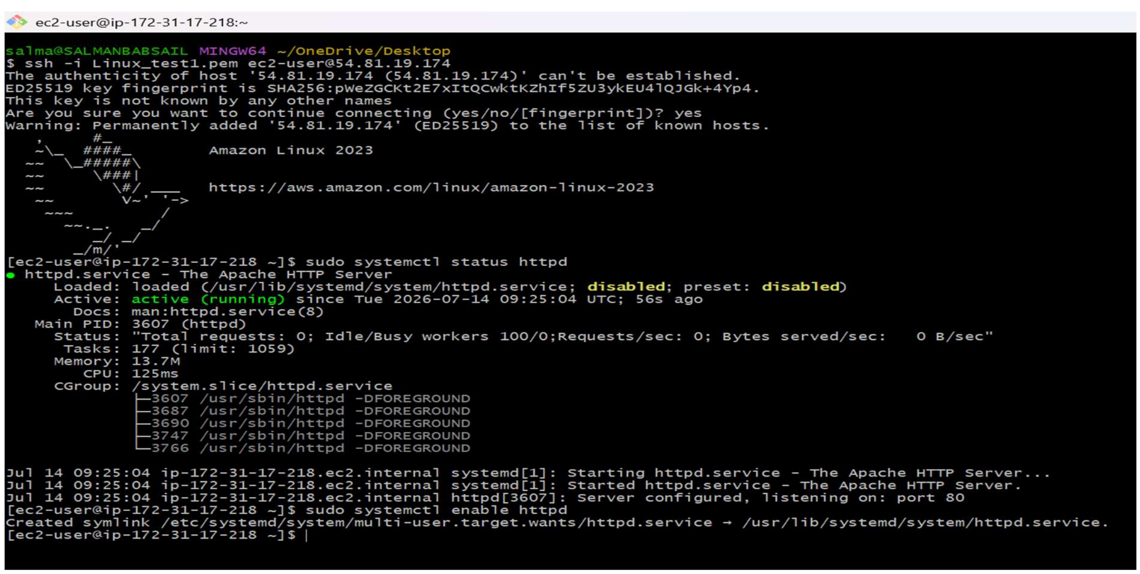
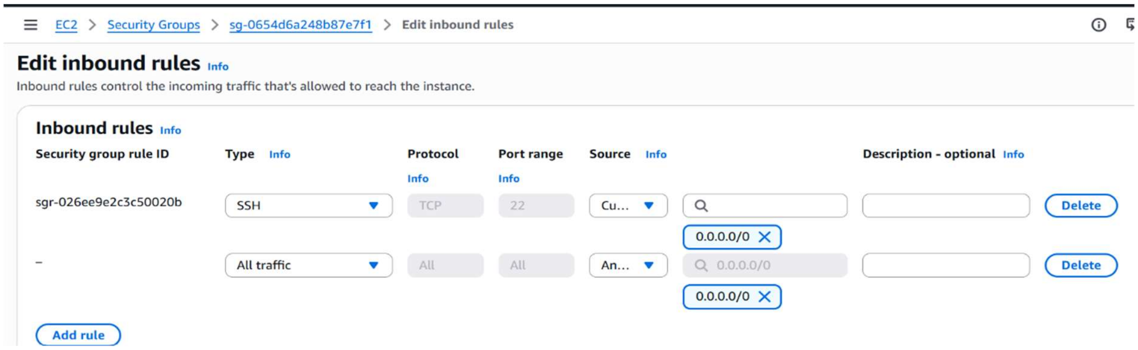
AWS -EC2 Task

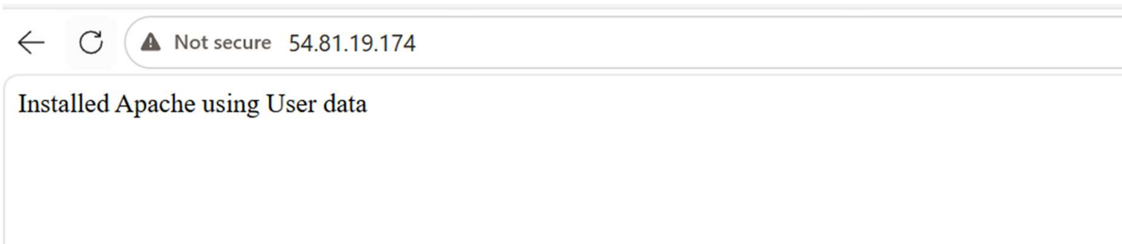
1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.



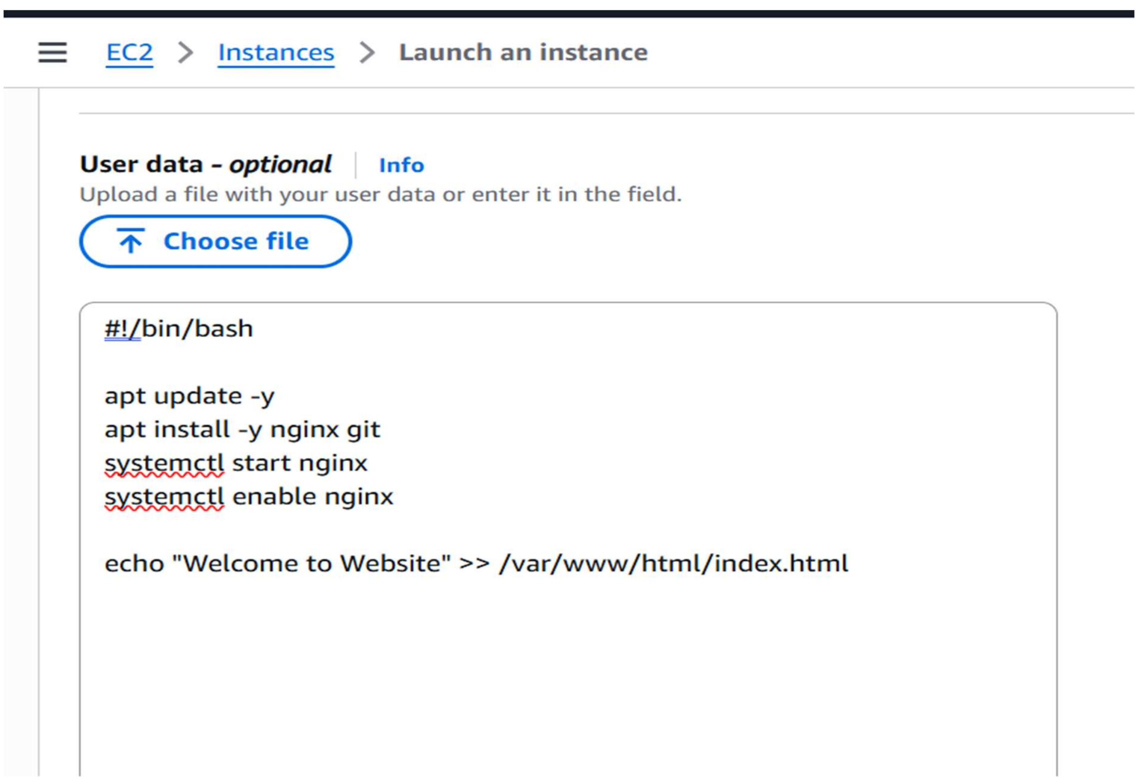


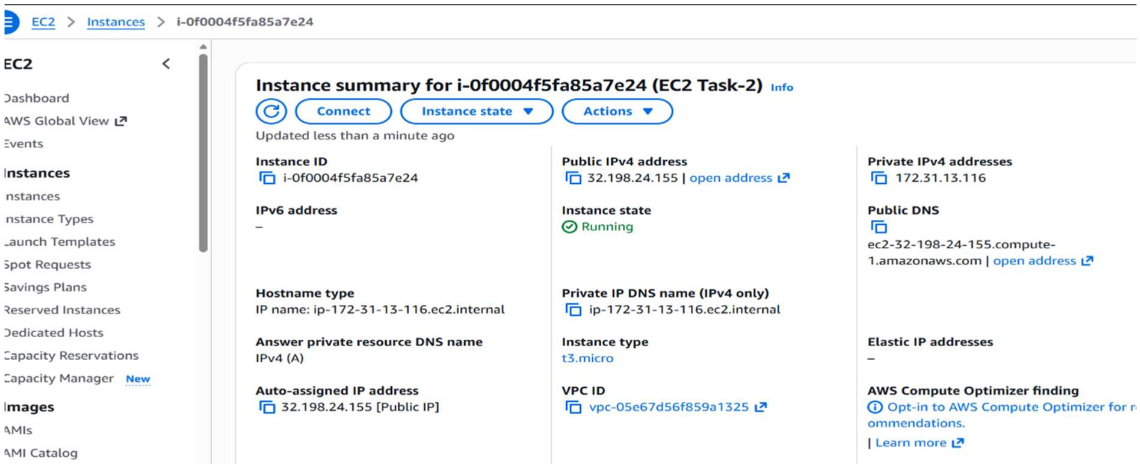


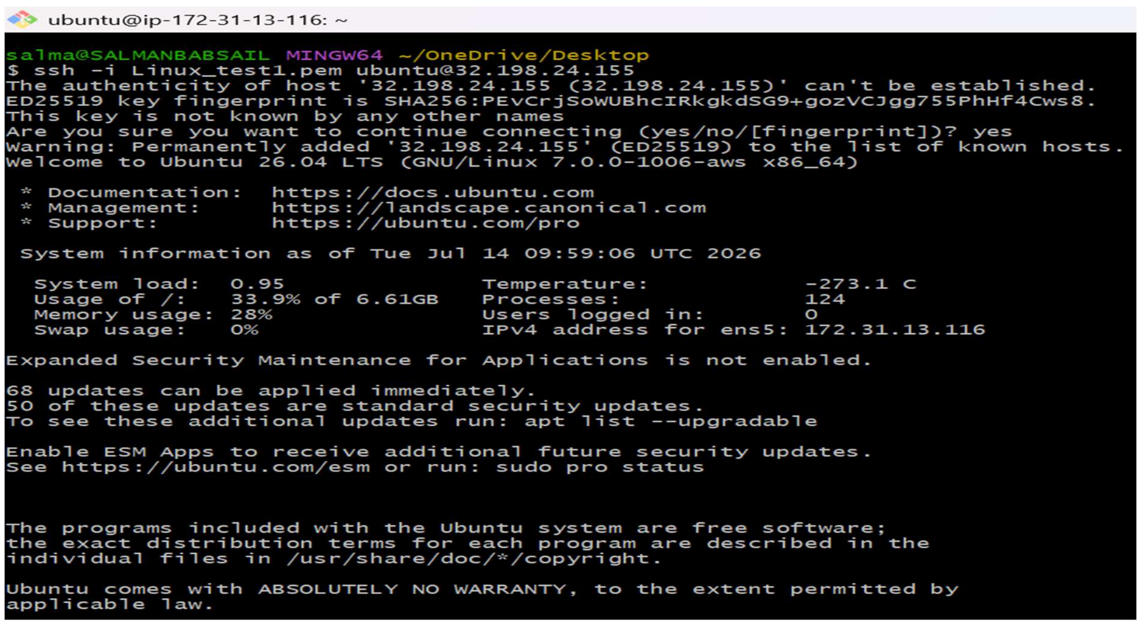


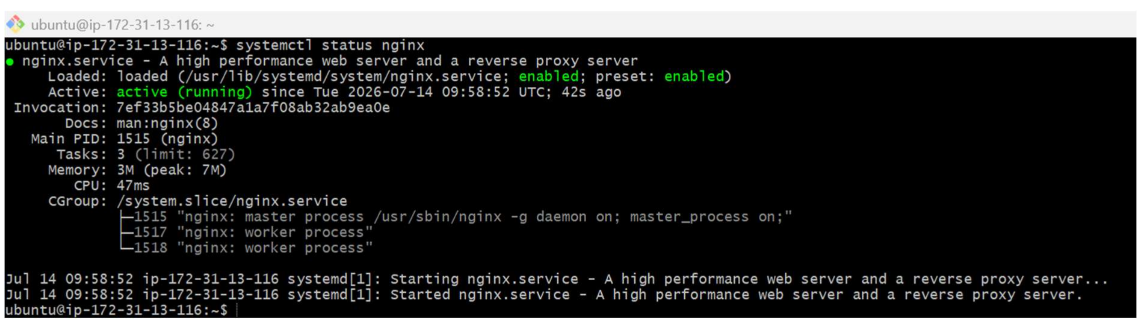


2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.



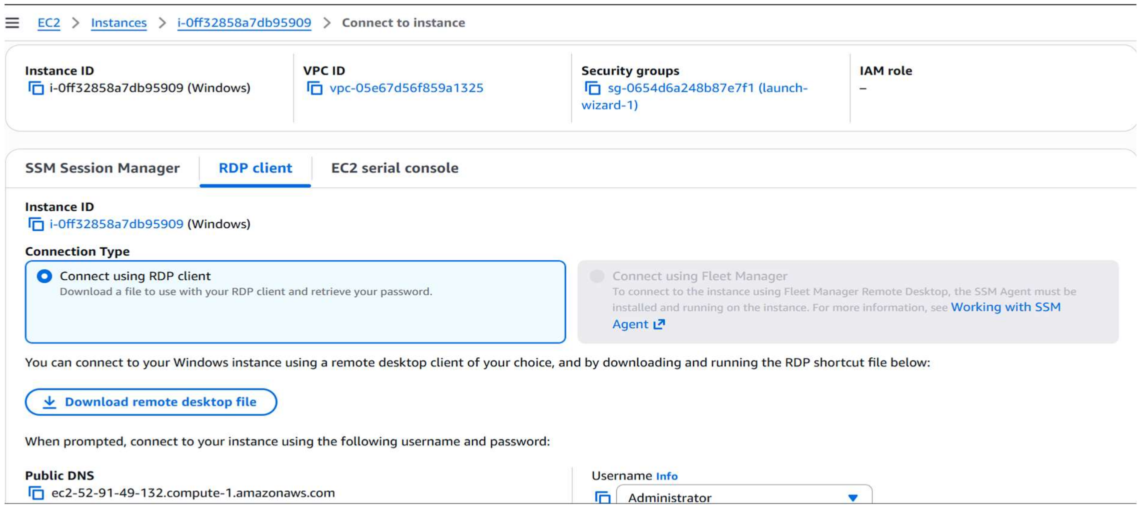

EC2 > Instances > i-0f0004f5fa85a7e24
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Instance summary for i-0f0004f5fa85a7e24 (EC2 Task-2) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-0f0004f5fa85a7e24
Public IPv4 address
32.198.24.155 | open address
Private IPv4 addresses
172.31.13.116
IPv6 address
–
Instance state
Running
Public DNS
ec2-32-198-24-155.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-13-116.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-13-116.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
32.198.24.155 [Public IP]
VPC ID
vpc-05e67d56f859a1325
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more

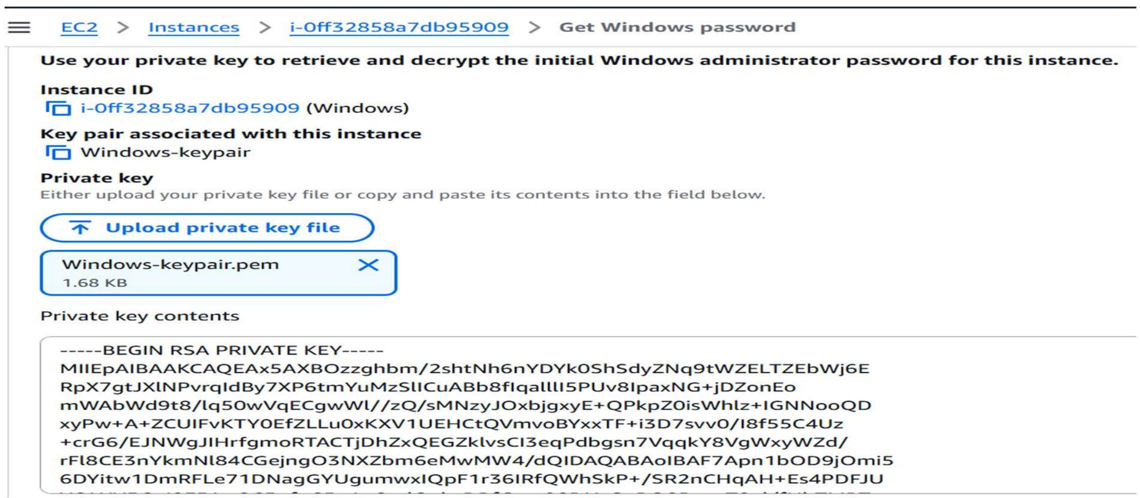

ubuntu@ip-172-31-13-116: ~
salma@SALMANBABSAIL MINGW64 ~/OneDrive/Desktop
$ ssh -i Linux_test1.pem ubuntu@32.198.24.155
The authenticity of host '32.198.24.155 (32.198.24.155)' can't be established.
ED25519 key fingerprint is SHA256:PEvCrjSowUBhcIRkgkdSG9+gozVCJgg755PhHf4Cws8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '32.198.24.155' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Tue Jul 14 09:59:06 UTC 2026
System load: 0.95
Temperature: -273.1 C
Usage of /: 33.9% of 6.61GB
Processes: 124
Memory usage: 28%
Users logged in: 0
Swap usage: 0%
IPv4 address for ens5: 172.31.13.116
Expanded Security Maintenance for Applications is not enabled.
68 updates can be applied immediately.
50 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

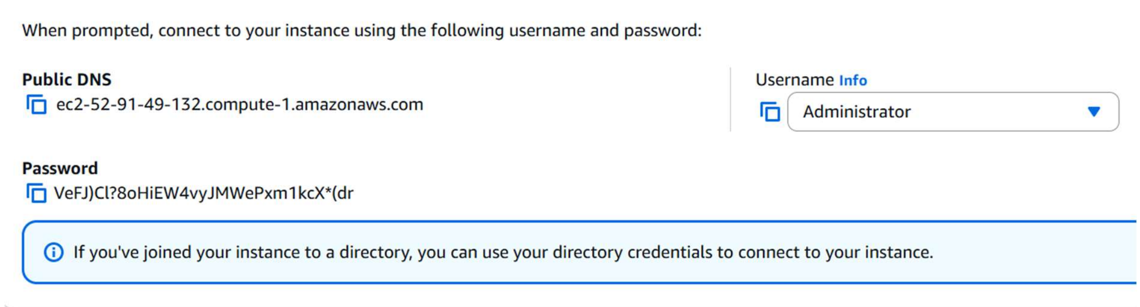

ubuntu@ip-172-31-13-116: ~
ubuntu@ip-172-31-13-116:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-07-14 09:58:52 UTC; 42s ago
Invocation: 7ef33b5be04847a1a7f08ab32ab9ea0e
Docs: man:nginx(8)
Main PID: 1515 (nginx)
Tasks: 3 (limit: 627)
Memory: 3M (peak: 7M)
CPU: 47ms
CGroup: /system.slice/nginx.service
├─1515 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─1517 "nginx: worker process"
└─1518 "nginx: worker process"
Jul 14 09:58:52 ip-172-31-13-116 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 14 09:58:52 ip-172-31-13-116 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-13-116:~$

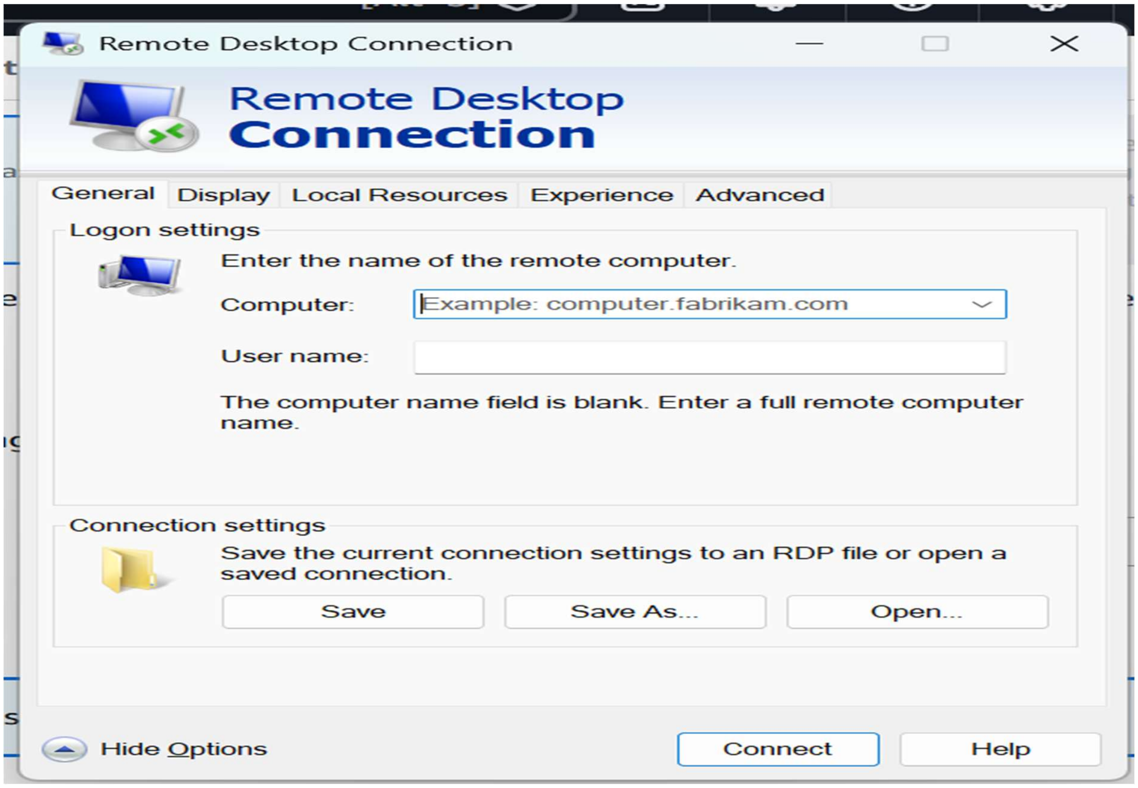

Not secure 32.198.24.155
Welcome to Website

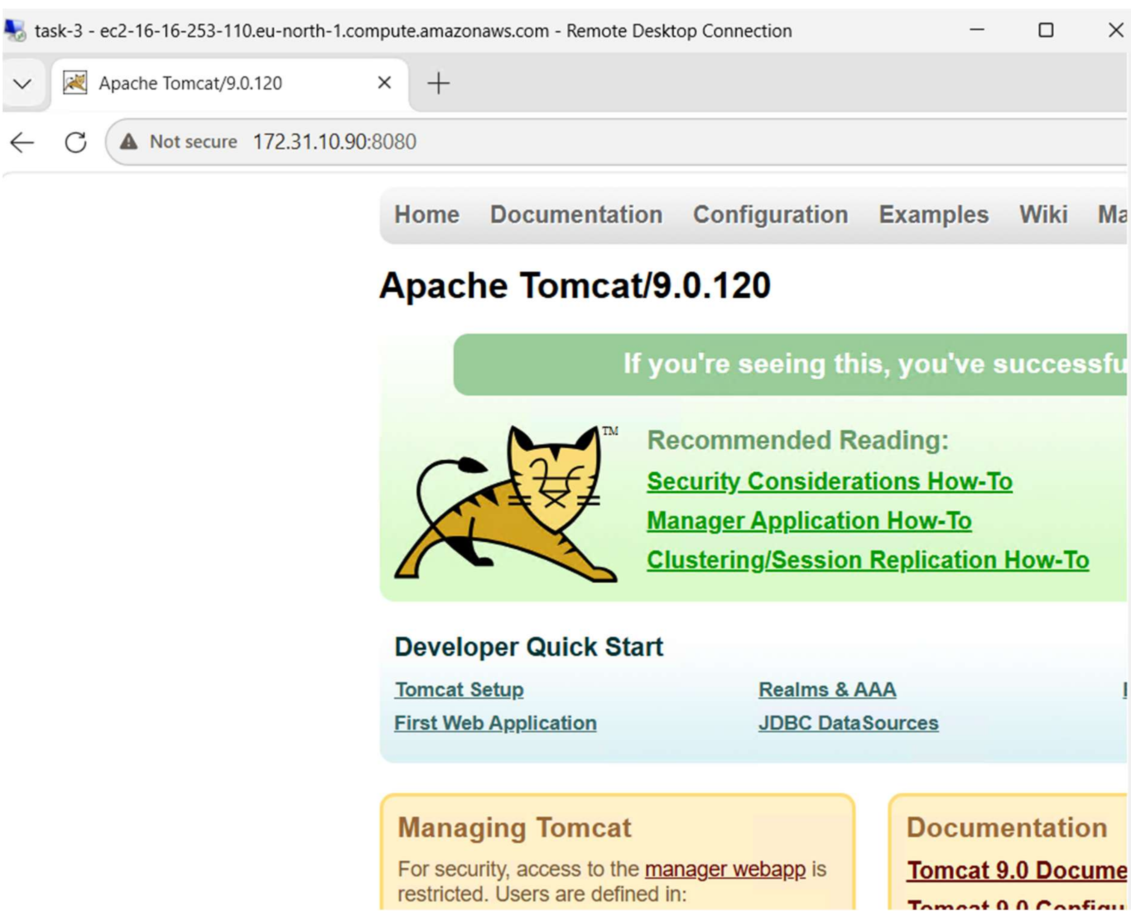
3. Launch one Windows server and install Tomcat on Windows.



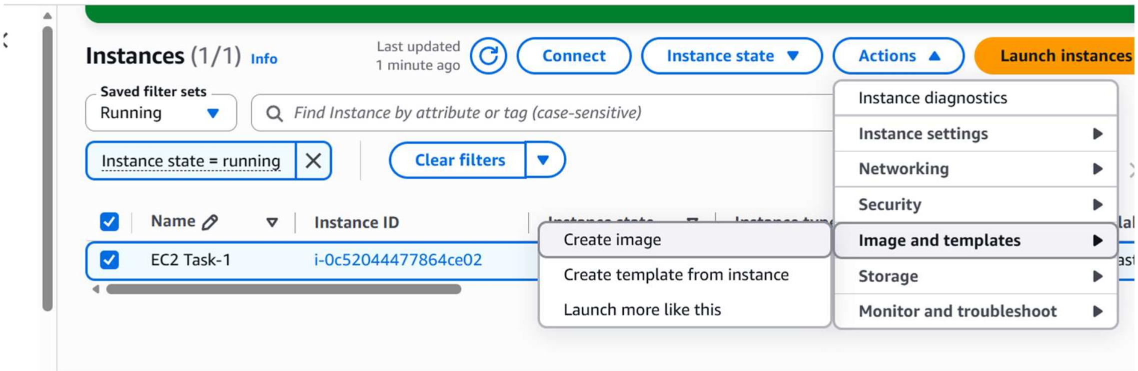


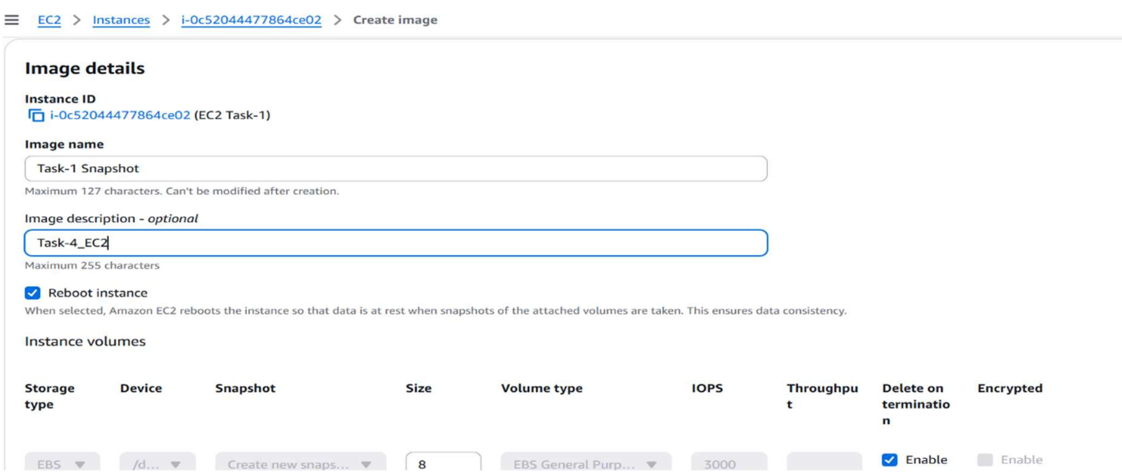


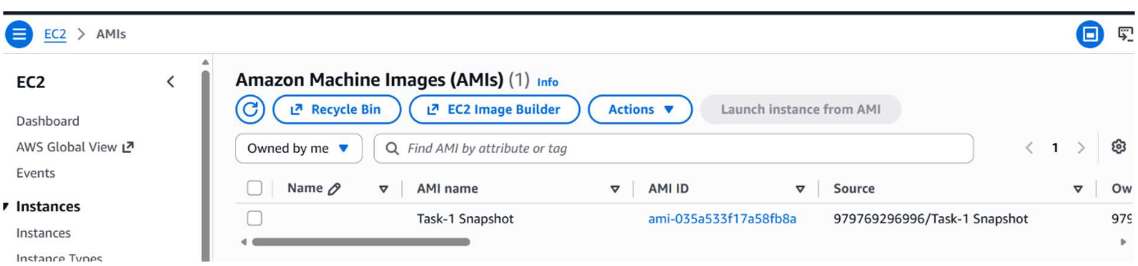

Remote Desktop Connection
Remote Desktop
Connection
General
Display
Local Resources
Experience
Advanced
Logon settings
Enter the name of the remote computer.
Computer:
Example: computer.fabrikam.com
User name:
The computer name field is blank. Enter a full remote computer name.
Connection settings
Save the current connection settings to an RDP file or open a saved connection.
Save
Save As...
Open...
Hide Options
Connect
Help

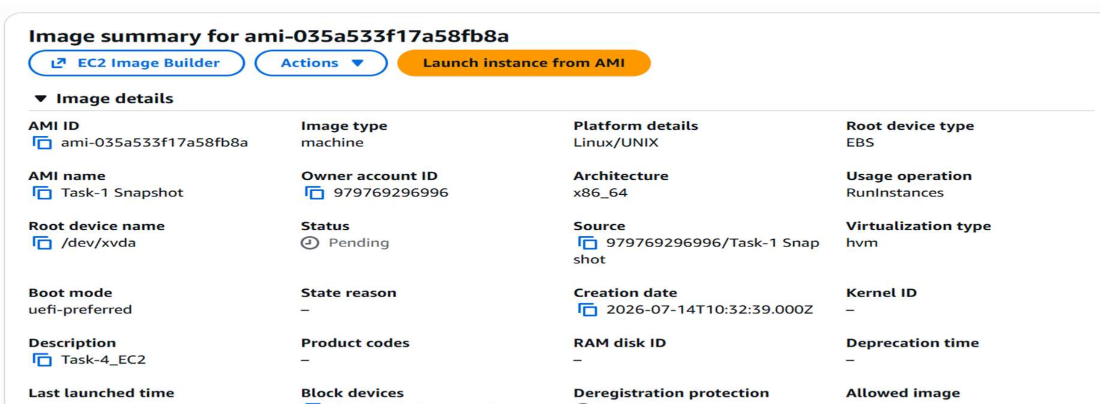

task-3 - ec2-16-16-253-110.eu-north-1.compute.amazonaws.com - Remote Desktop Connection
Apache Tomcat/9.0.120
Not secure 172.31.10.90:8080
Home
Documentation
Configuration
Examples
Wiki
Apache Tomcat/9.0.120
Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To
Developer Quick Start
Tomcat Setup
Realms & AAA
First Web Application
JDBC DataSources
Managing Tomcat
For security, access to the manager webapp is restricted. Users are defined in:
Documentation

4. Take a snapshot of the instance created in Task 1.









5. Assign passwordless authentication for the EC2 created in Task 2.

ubuntu@ip-172-31-13-116: ~/.ssh

```
ubuntu@ip-172-31-13-116:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:wsK8FZbWN7HVpPFGtOuUg1INGtevPhgvc3BCMZIwK6Q ubuntu@ip-172-31-13-116
The key's randomart image is:
+--[ED25519 256]--+
|      . o..o+*B.o|
|     o o oo=**o*.|
|    E * o +oooooo|
|   o + o . .... o|
|    + + S  .   o |
|     + .    + o  |
|    .       O    |
|          + =    |
|           + .   |
+----[SHA256]-----+
ubuntu@ip-172-31-13-116:~$ cat /home/ubuntu/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILv2cyBILqpJJiCw/+1F/yohjS9P73SGgoOLBgTCggRS ubuntu@ip-172-31-13-116
ubuntu@ip-172-31-13-116:~$ cd .ssh/
ubuntu@ip-172-31-13-116:~/.ssh$ vi authorized_keys
```

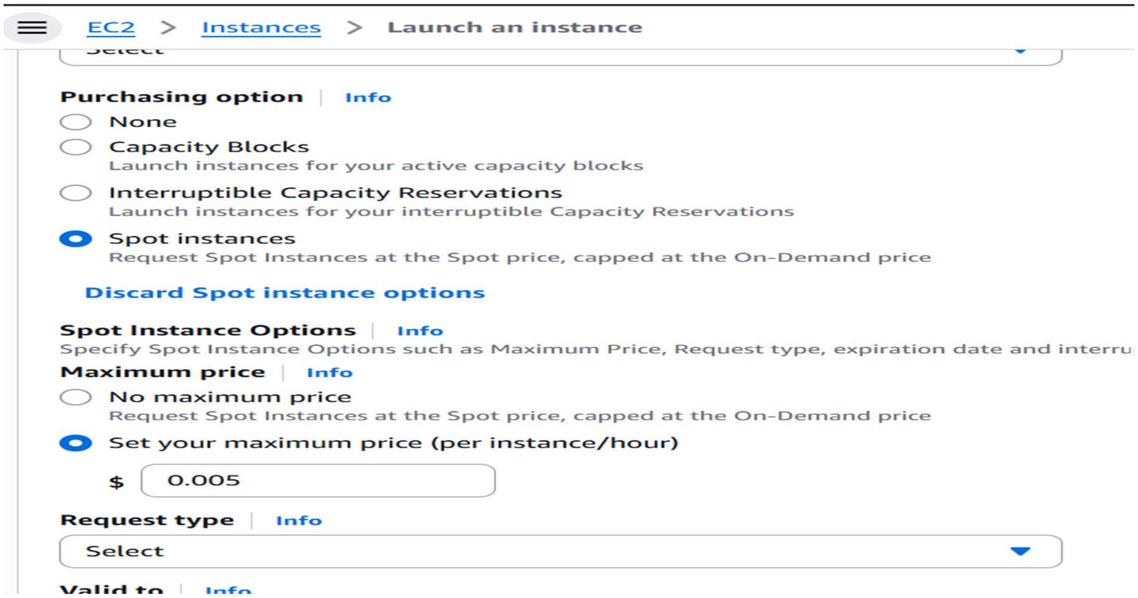
ubuntu@ip-172-31-13-116: ~/.ssh

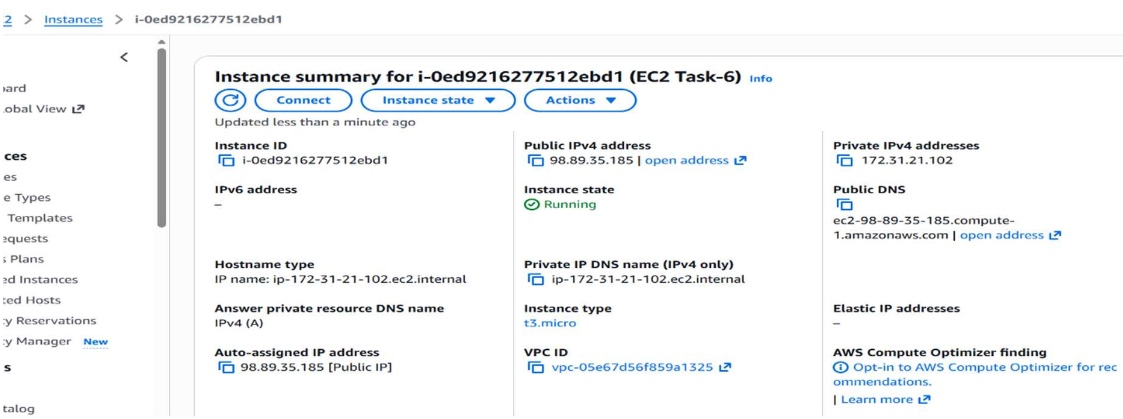
```
ubuntu@ip-172-31-13-116:~$ cd .ssh/
ubuntu@ip-172-31-13-116:~/.ssh$ ls
authorized_keys  id_ed25519  id_ed25519.pub  known_hosts
ubuntu@ip-172-31-13-116:~/.ssh$ cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILv2cyBILqpJJiCw/+1F/yohjS9P73SGgoOLBgTCggRS ubuntu@ip-172-31-13-116


ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDJOHkTZuVzpQHYxbXrzRovAqVf5xrjEhMz9u9VDhZiAL9Dero6FWxUTLD1yV3IgEq7
r6SxT5CymKQvYYp3SvWKjOj1njqyiT4HG3GHbOdb6tQX2VYxs4MLJodIsCoYjLnIn3LHUlckyVXZJ/udfAbnbRIF7gzG7EE8ROPd64fl
bMLUfLM5ADRmP8hxJGdgcOoXATI54V7r/3Cj Linux_test1
ubuntu@ip-172-31-13-116:~/.ssh$
```

ubuntu@ip-172-31-13-116: ~

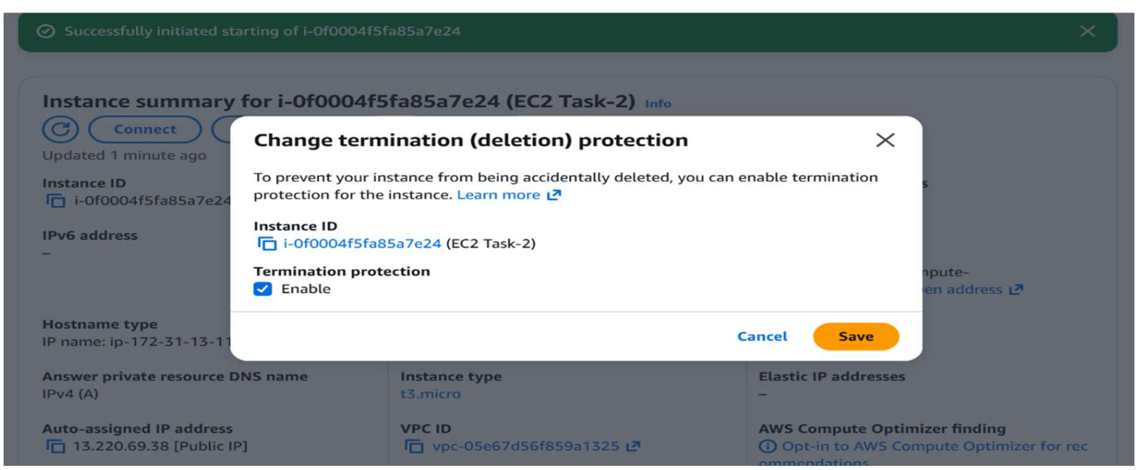
```
ubuntu@ip-172-31-13-116:~/.ssh$ ssh ubuntu@3.238.2.151
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Jul 14 11:09:48 UTC 2026

  System load:  0.0                Temperature:           -273.1 C
  Usage of /:   34.2% of 6.61GB    Processes:             121
  Memory usage: 27%                Users logged in:       0
  Swap usage:   0%                 IPv4 address for ens5: 172.31.13.116


Expanded Security Maintenance for Applications is not enabled.

68 updates can be applied immediately.
50 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Tue Jul 14 10:36:42 2026 from 103.203.172.230
ubuntu@ip-172-31-13-116:~$
```

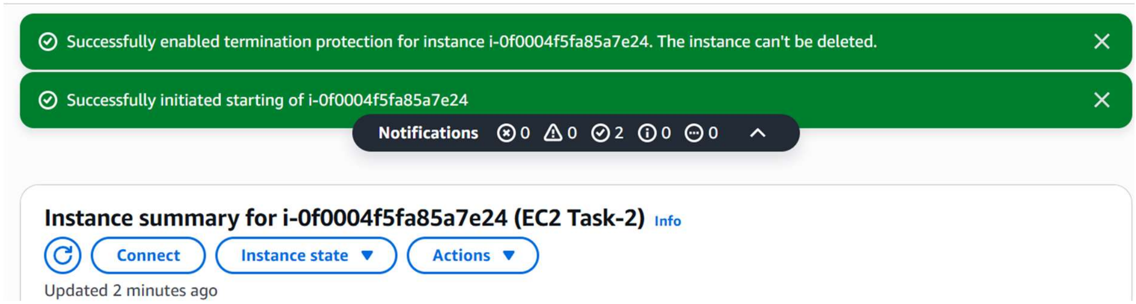
6. Launch any EC2 using the spot purchasing option.



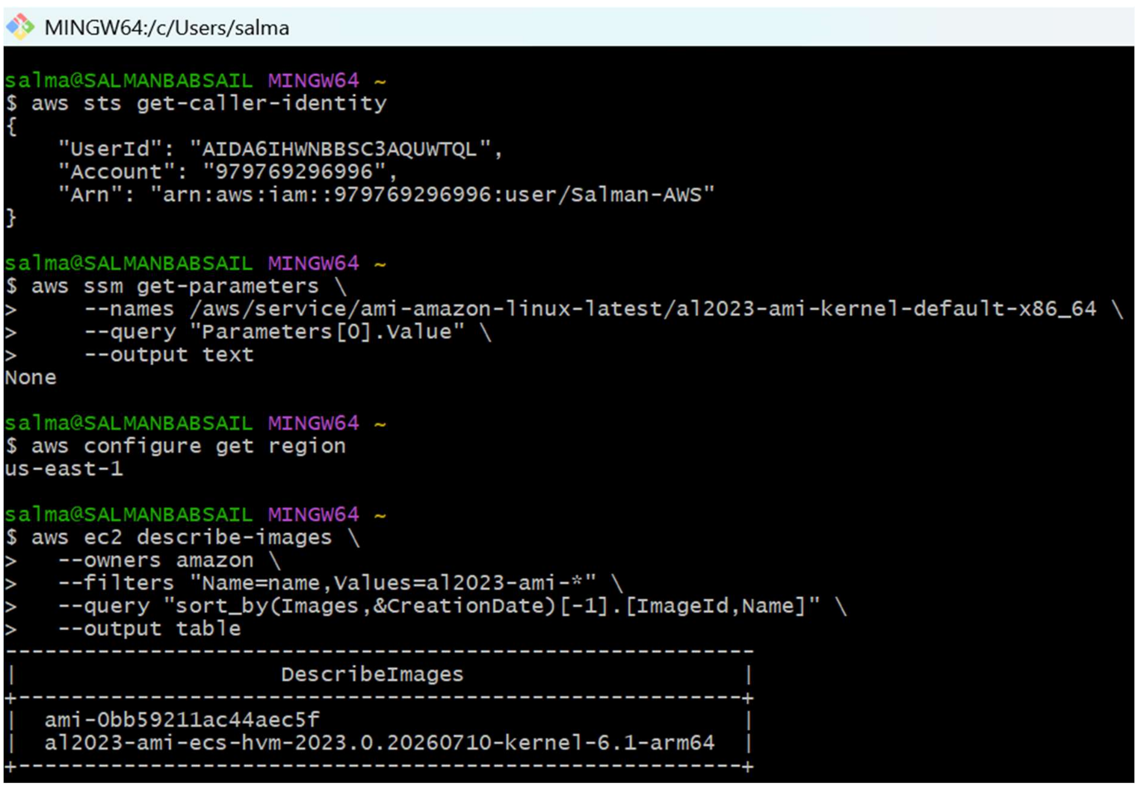


7. Enable termination policy on the EC2 created in Task 2.





8. Launch one EC2 using AWS CLI.



MINGW64:/c/Users/salma

```
salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 describe-images \
>   --owners amazon \
>   --filters \
>     "Name=name,Values=al2023-ami-2023*-kernel-6.1-x86_64" \
>     "Name=state,Values=available" \
>   --query "sort_by(Images,&CreationDate)[-1].[ImageId,Name]" \
>   --output table
----------------------------------------------------
|                  DescribeImages                  |
+--------------------------------------------------+
|  ami-0fd6240f599091088                           |
|  al2023-ami-2023.12.20260710.0-kernel-6.1-x86_64 |
+--------------------------------------------------+

salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 describe-key-pairs \
>   --query "KeyPairs[*].KeyName" \
>   --output table
--------------------
| DescribeKeyPairs |
+------------------+
|  windows-keypair |
|  Linux_test1     |
+------------------+

salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 describe-security-groups \
>   --query "SecurityGroups[*].[GroupId,GroupName]" \
>   --output table
-------------------------------------------------
|            DescribeSecurityGroups             |
+-----------------------+-----------------------+
|  sg-0af8afa7d8f2192f1 |  default              |
|  sg-0654d6a248b87e7f1 |  launch-wizard-1      |
+-----------------------+-----------------------+

salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 describe-subnets \
>   --query "Subnets[*].[SubnetId,AvailabilityZone]" \
>   --output table
----------------------------------------------
|              DescribeSubnets               |
+---------------------------+----------------+
|  subnet-05ebc072052c8de40 |  us-east-1c    |
|  subnet-03388cfb3ecca111f |  us-east-1b    |
|  subnet-0a005cceeebb23e47 |  us-east-1e    |
|  subnet-08c419a8c502d6ba4 |  us-east-1d    |
|  subnet-09cb7b2ed39680aea |  us-east-1a    |
|  subnet-0a673039be6c30c4d |  us-east-1f    |
+---------------------------+----------------+
```

MINGW64:/c/Users/salma

```
salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 run-instances \
>   --image-id ami-0fd6240f599091088 \
>   --instance-type t3.micro \
>   --key-name Linux_test1 \
>   --security-group-ids sg-0654d6a248b87e7f1 \
>   --subnet-id subnet-09cb7b2ed39680aea \
>   --associate-public-ip-address \
>   --count 1 \
>   --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=CLI-EC2}]'
{
    "ReservationId": "r-03993a3eed4eefc33",
    "OwnerId": "979769296996",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "fd8eb462-bd6a-4ab3-88f5-fbd0e9e66f52",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2026-07-15T10:07:50+00:00",
                        "AttachmentId": "eni-attach-04faaca28c4007153",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
                        {
                            "GroupId": "sg-0654d6a248b87e7f1",
                            "GroupName": "launch-wizard-1"
                        }
                    ],
                    "Ipv6Addresses": [],
                    "MacAddress": "02:4b:d6:70:bf:5f",
                    "NetworkInterfaceId": "eni-07ad285eb2e23c830",
                    "OwnerId": "979769296996",
                    "PrivateDnsName": "ip-172-31-3-102.ec2.internal",
                    "PrivateIpAddress": "172.31.3.102",
                    "PrivateIpAddresses": [
                        {
                            "Primary": true,
                            "PrivateDnsName": "ip-172-31-3-102.ec2.internal",
                            "PrivateIpAddress": "172.31.3.102"
                        }
                    ],
                    "SourceDestCheck": true,
                    "Status": "in-use",
```

```
            "RebootMigration": "default"
        },
        "CurrentInstanceBootMode": "uefi",
        "Operator": {
            "Managed": false
        },
        "InstanceId": "i-0f56f7749ec160661",
        "ImageId": "ami-0fd6240f599091088",
        "State": {
            "Code": 0,
            "Name": "pending"
        },
        "PrivateDnsName": "ip-172-31-3-102.ec2.internal",
        "PublicDnsName": "",
        "StateTransitionReason": "",
        "KeyName": "Linux_test1",
        "AmiLaunchIndex": 0,
        "ProductCodes": [],
        "InstanceType": "t3.micro",
        "LaunchTime": "2026-07-15T10:07:50+00:00",
        "Placement": {
            "AvailabilityZoneId": "use1-az1",
            "GroupName": "",
            "Tenancy": "default",
            "AvailabilityZone": "us-east-1a"
        },
        "Monitoring": {
            "State": "disabled"
        },
        "SubnetId": "subnet-09cb7b2ed39680aea",
        "VpcId": "vpc-05e67d56f859a1325",
        "PrivateIpAddress": "172.31.3.102"
}
```

```
salma@SALMANBABSAIL MINGW64 ~
$ aws ec2 describe-instances \
>   --query "Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress,Tags[0].Value]" \
>   --output table
---------------------------------------------------------------------------------
|                              DescribeInstances                                |
+----------------------+----------+----------------+-------------------+
|  i-0d1635319dfe50f59 |  stopped |  None          |  Linux_Commands   |
|  i-05f28a37e21fa5724 |  stopped |  None          |  Amazon Linux     |
|  i-0ff32858a7db95909 |  stopped |  None          |  Windows          |
|  i-0f56f7749ec160661 |  running |  3.238.194.139 |  CLI-EC2          |
|  i-0f0004f5fa85a7e24 |  stopped |  None          |  EC2 Task-2       |
|  i-0c52044477864ce02 |  stopped |  None          |  EC2 Task-1       |
+----------------------+----------+----------------+-------------------+
```

```
salma@SALMANBABSAIL MINGW64 ~/OneDrive/Desktop
$ chmod 400 Linux_test1.pem

salma@SALMANBABSAIL MINGW64 ~/OneDrive/Desktop
$ ssh -i Linux_test1.pem ec2-user@3.238.194.139
The authenticity of host '3.238.194.139 (3.238.194.139)' can't be established.
ED25519 key fingerprint is SHA256:6VOT3Xtbw0lYIL+9WXiHY7ZdBNDV5wyOQRjVQ04qAlQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.238.194.139' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-3-102 ~]$ 
```